**1. Pair Counting**

Given an integer n, count how many pairs (i, j) exist such that $1 \le i < j \le n$.

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**2. Print All Pairs**

Given n, print all ordered pairs (i, j) such that $1 \le i, j \le n$.

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**3. Equal Elements**

Given an array of size n, count how many pairs (i, j) exist such that $i < j$ and $a[i] = a[j]$.

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**4. Subarray Sum Listing**

Given an array, print the sum of all possible subarrays.

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**5. Maximum Pair Sum**

Given an array, find the maximum value of $a[i] + a[j]$ for $i < j$.

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**6. Divisible Pairs**

Given an array, count pairs (i, j) such that $i \neq j$ and $a[i] \% a[j] = 0$.

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**7. Duplicate Check**

Given an array, determine if any element appears more than once.

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**Medium Level**

**8. Two Sum (All Pairs)**

Given an array and integer k, print all pairs (i, j) such that $a[i] + a[j] = k$.

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**9. Inversion Count**

Count pairs (i, j) such that $i < j$ and $a[i] > a[j]$.

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**10. Triplet Sum**

Find all triplets (i, j, k) such that $i < j < k$ and a[i] + a[j] + a[k] = target.

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**11. Subarrays with Sum K**

Count the number of subarrays whose sum equals k.

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**12. Row Sum in Matrix**

Given a matrix, find the sum of each row.

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**13. Matrix Transpose**

Given a matrix, output its transpose.

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**14. Matrix Multiplication**

Given two matrices, compute their product.

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□ **Hard Level (15–20)**

**15. Maximum Subarray Sum (Brute Force)**

Find the maximum sum of any subarray using nested loops.

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**16. Count Rectangles in Grid**

Given an $n \times m$ grid, count total rectangles that can be formed.

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**17. Palindrome Substrings**

Given a string, count all substrings that are palindromes.

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**18. Pair Frequency**

Given an array, count how many times each ordered pair (a[i], a[j]) appears.

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### 19. Valid Triangle Count

Given an array, count triplets that can form a valid triangle.

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### 20. Submatrix Sum

Given a matrix, calculate the sum of all possible submatrices.

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### Bonus

### 21. Count Quadruples

Find (i, j, k, l) such that $i < j < k < l$ and sum equals target.

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### 22. Distinct Subarrays

Count subarrays with all distinct elements (brute force).

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### 23. Equal XOR Subarrays

Count subarrays where XOR of elements is 0 (brute force).